

Anas Shajahan

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING
B.TECH DEGREE SECOND SEMESTER FIRST INTERNAL EXAMINATION
APRIL 2024

23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++
(S2-EC A)
(2023 Scheme)

Time:2 Hours

Maximum marks:30

Course outcomes:

1. To understand basic concepts of Object – Oriented Programming.
2. Describe object - oriented paradigm with concepts of streams, classes, functions, data and objects.
3. To implement object- oriented programming constructs like encapsulation, inheritance and polymorphism.
4. To understand dynamic memory management techniques using pointers, constructors, destructors etc.
5. To identify classes including data, methods and the relationship among classes from a given problem statement and solve the problem using object -oriented constructs in C++

Part A

(Answer all questions)

Marks (5×2=10)

I.

- 1) Compare and contrast OOP languages with procedure oriented languages. (CO1, L2, PO-1,12)
- 2) Differentiate formatted and unformatted data. (CO2, L2, PO-1,2,12)
- 3) WAP to display A to Z characters using while loop. (CO3, L3, PO-1,2,3,12)
- 4) Explain the uses of private and public keywords. How are they different from each other?(CO5, L2, PO-1,2,3,5,6,8,12)
- 5) WAP to declare a class with three integer public data variables. Initialize and display them. (CO5, L3, PO-1,2,3,5,6,8,12)

Part B

II.

Marks (2×10=20)

1. (a) WAP to define overloaded function add() for integer and float and perform the addition. (CO3, L3, PO-1,2,3,12) (2.5)
- (b) WAP to check whether entered string is a palindrome or not. (CO3, L3, PO-1,2,3,12) (2.5)
- (c) Explain the key concepts of object oriented programming. (CO1, L1, PO-1,12) (5)

OR

- 2.(a) WAP to calculate the factorial of a number N. User enters the value of N. (CO3, L3, PO-1,2,3,12) (2.5)

(b) WAP to define function cube() as inline for calculating cube(CO3, L3, PO-1,2,3,12) (2.5)

(c) Explain the three methods by which we can pass value to the function. Demonstrate each method with suitable program. (CO2, L2, PO-1,2,12) (5)

3.(a) WAP to define three classes. Define a friend function. Read and display the data for three classes using common functions and friend function. (CO2, L3, PO-1,2,12) (5)

(b) What are static member variables and functions? How they differ from other member variable and functions? How static variables are initialized? How are static functions invoked? (CO2, L2, PO-1,2,12) (5)

OR

4. (a) What are constructors and destructors? Explain their characteristics. Explain types of constructors. (CO4, L2, PO-1,12) (5)

(b) WAP to declare a class with three data members. Declare overloaded constructors with no arguments, one argument, two arguments and three arguments. Pass values in the object declaration statement. Create 4 objects and pass values in such a way that the entire 4 constructors are executed one by one. Write appropriate messages in constructor so that the execution of program can be understood. (CO2, L3, PO-1,2,12) (5)

(L1-10%, L2-42%, L3-48%)

>>>><<<<